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April 21, 1975
INTERIM REVISION OF
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February 11, 1958

INTERIM FEDERAL SPECIFICATION

LAMP, INCANDESCENT, (ELECTRIC, MINIATURE AND SUBMINIATURE, TUNGSTEN-FILAMENT)

(GENERAL SPECIFICATION)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Mechanical and Electrical Division, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification W-L-111C, dated February 11, 1958.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers miniature and subminiature incandescent tungsten-filament electric lamps. For purposes of this specification, miniature and subminiature lamps are defined as follows: Miniature lamps have bulb diameters larger than 0.250 inch and usually equipped with bases smaller than those used on large incandescent lamps as covered by W-L-101. Subminiature lamps have bulb diameters of 0.250 and less, and are generally rated at 28 volts or less. These lamps are designed for automotive, flashlight, aviation, indicator, and similar service.

1.2 Classification. Lamps shall be as indicated on the Detail Specification Sheets.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-676 - Boxes, Setup.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood Cleated-Plywood.
PPP-B-621 - Boxes, Nailed and Lock-Corner.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specifications:

MIL-L-10547 - Liners, Case, Waterproof.
MIL-C-25050A - Color, Aeronautical Lights and Lighting Equipment, General Requirements For.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI):

C78.390 - Designation of Miniature Incandescent Lamps.

(Copies of the ANSI Standards may be obtained from 1430 Broadway, New York, NY 10018.)

Society of Automotive Engineers, Inc.:

SAE J 573 - Lamp Bulbs and Sealed Units.

(Copies of the Society of Automotive Engineers, Inc., standards may be obtained from Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15086.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Quality control and production. The manufacturer shall have a documented quality control system, approved by the Government. Complete and accurate records of the inspections performed in accordance with this system shall be made available to the Government for a period of at least the last 2 years.

3.2 Qualification. Lamps furnished under this specification shall be a product which has been tested in accordance with the qualification tests described in section 4 of this specification and have been listed on or approved for listing on the applicable Federal Qualified Product List.

3.3 Materials.

3.3.1 Metals. When specified (see slash sheets), metals shall be corrosion resistant or suitably treated to resist corrosion caused by fuels, salt spray, or atmospheric conditions as may be encountered in storage or normal service.

3.4 Design and construction.

3.4.1 The lamps shall be designed in accordance with the applicable detail specification sheet. The lamp shall be constructed in accordance with standard commercial practice, except for changes necessary to provide lamps that will conform to this specification.

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3.4.2 Color. The detail specification sheet shall specify if a lamp is to be a color and what color.

3.5 Performance.

3.5.1 Life. The life requirement for lamps furnished under this specification shall be either average laboratory life or minimum laboratory life, as specified on the detail specification. Where average life is specified, lamps shall give an average life to burn out of not less than that specified in the detail specification as modified by paragraph 4.6.2.1. Where minimum life is specified, lamps shall give the minimum life as specified on the detail specification with no more than 1 percent failures (burnouts). Where sample sizes are such that 1 percent failures result in less than one lamp, a single failure would be permitted.

3.5.2 Current. Lamps furnished under this specification shall draw current initially, at a rate within the range specified in the applicable detail specification, when tested in accordance with 4.6.1.3.

3.5.3 Initial candlepower. Lamps furnished under this specification shall have an initial candlepower output within the range specified in the applicable detail specification sheet when tested as specified in 4.6.1.1.

3.5.4 Candlepower maintenance. The candlepower maintenance for lamps furnished under this specification shall be as specified in the detail specification sheet when tested in accordance with paragraph 4.6.1.4.

3.5.5 Efficacy. The initial efficacy (spherical candles per watt) for lamps furnished under this specification shall be as specified in the detail specification sheet when tested as specified in 4.6.1.2.

3.6 Dimensions. Bulb, base, and filament dimensions, and axial alignment as applicable, shall be as specified on the applicable detail specification sheet. All dimensions shown on the drawings on the detail specification sheets are in inches.

3.7 Marking. Except where space does not permit marking on the bulb or base, each lamp shall be marked on the bulb or base with the trade number of the lamp and the manufacturer's name or trademark.

3.8 Workmanship. Each lamp shall be fabricated and furnished in a thoroughly workmanlike manner. Each lamp shall be free from blemishes and defects. Marking shall be clear, legible, and durable. Soldering, welding, brazing, cementing, and wiring shall be thorough. Alignment of parts shall be accurate. Each lamp shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, and other foreign material removed and shall be free of any defects which will affect form, fit, or function.

3.9 Vibration. When specified in the detail specification, and when tested in accordance with 4.6.3, the lamp shall pass the vibration test. Damage or loosening of any part of any lamp by this test shall be cause for rejection and the lamp type shall fail to qualify.

3.10 Shock. When specified in the detail specification, and when tested in accordance with 4.6.4, the lamps shall pass the shock test. Damage or loosening of any part of any lamp by this test shall be cause for rejection and the lamp type shall fail to qualify.

3.11 Humidity. When specified in the detail specification, and when tested in accordance with 4.6.5, the lamps shall pass the humidity test.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Classification. The examination and testing of lamps shall be classified as:

(a) Qualification.
(b) Quality conformance.

4.3 Qualification.

4.3.1 Qualification tests and inspections. Qualification tests and inspections shall consist of all tests and inspections specified in 4.6.

4.3.2 Qualification sample. The size of the qualification test sample and the sampling procedure will be as specified by the Standards divisions, Federal Supply Service, General Services Administration.

4.3.3 Failure to qualify. Lamps failing to comply with any of the performance requirements and test of 3.5.1, 3.5.2, 3.5.3, 3.5.4, 3.5.5, 3.9, 3.10, 3.11, 4.6.1, 4.6.1.1, 4.6.1.2, 4.6.1.3, 4.6.1.4, 4.6.2, 4.6.3, 4.6.4, and 4.6.5 shall fail to qualify.

4.4 Definitions.

4.4.1 Light center for C-2 shaped filaments. Light center is a point lying in a plane through the axis of the filament coils midway between the "top" and "bottom" of the lighted filament and midway between the outer edges of the filament coils at the horizontal level of the light center.

NOTE: Filament top is the upper extremity of the highest lighted turn (away from base) in either coil segment. Filament bottom is the underside of the lowest lighted turn (toward base) in either coil segment. The lowest lighted turn is considered for this purpose as being the first turn above the first inside angle between turns of the projected filament image.

4.4.2 Light center for C-6 shaped filaments. Light center is a point lying in the axis of the filament coil midway between the end turns of the lighted filament.

4.4.3 Light center length. Light center length is the distance from light center to the specified reference point on the lamp.

4.4.3.1 Reference points on the lamp base for light center length measurement.

Screw base - End of the center contact.
Bayonet base - Top of base pins.
Single contact or double contact prefocus - Bottom of indentations on prefocusing collar.
Single contact miniature flanged - Top of bosses in flange.
Double contact index - Top of base pin nearer bottom contacts.
SC midget flange - Top of flange.
Submidget flange - Top of flange.

4.4.4 Axial alignment. Axial alignment for automotive lights shall be as shown in SAE STD J573 for each specific lamp.

4.5 Quality conformance provisions.

4.5.1 Acceptance inspection.

4.5.1.1 Sampling for acceptance inspection. Unless otherwise specified, minimum sampling shall be performed in accordance with the provisions set forth in MIL-STD-105, inspection level S-3.

4.5.1.2 Inspection lot. Inspection lot shall consist of lamps of one designation and size, from an identifiable production period, from one manufacturer, and one plant.

4.5.1.3 In-process inspection. Inspections shall be performed throughout the manufacturing process to assure that lamps produced meet the requirements of this specification.

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4.5.1.4 Acceptance tests. Each month during the contract, while the lamps are in the process of manufacture, each specified lamp type shall be tested to the photometric requirements (see 3.5.3, 3.5.4, and 3.5.5), power consumption requirements (see 3.5.2), and dimensional requirements (see 3.5.6) of this specification. Samples for acceptance testing shall be selected for acceptance testing shall be selected in accordance with MIL-STD-105D, inspection level S-3 (minimum acceptable sample level), with acceptance criteria as stated in the applicable detail specification and as stated in paragraph 4.6.6. If a lamp fails to comply with any of the above referenced requirements, it shall be considered defective. In case of test failure, the contractor shall suspend offering lamps of the designation until corrective measures, as necessary, have been taken and another sample of at least 32 lamps has been tested and passed.

NOTE: When the acceptance criteria results in a fraction of a lamp or a whole number and a fraction, the next higher whole number shall be used for acceptance.

4.5.1.5 Visual examination and life testing. Visual examination and life testing shall be a requirement of the manufacturer's quality control system (see 3.1). Assuming the quality control system is acceptable to the Government, no additional visual examination or life testing will be required for acceptance except as might be conducted under the provisions of 4.1.

4.5.1.6 Examination of preparation for delivery. An examination shall be made to determine that the packaging, packing, and marking comply with section 5 of this specification. Defects shall be scored in accordance with table III. The sample unit shall be one shipping container fully prepared for delivery selected at random just prior to the closing operation. Defects of closure shall be examined on closed shipping containers. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level of 4.0 expressed in terms of percent defective.

TABLE III. Classification of preparation for delivery defects

Examine	Defects
Marking (exterior and interior container)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components such as incomplete closure of case liners of container flaps, loose strapping, inadequate stapling; bulging or distortion of containers.

4.6 Tests and examinations.

4.6.1 Photometric and electrical tests. Photometric and electrical tests shall be conducted as specified herein and in accordance with the procedures set forth by the Illuminating Engineering Society (IES), reference the IES Handbook 5th Edition. The voltage for photometry shall be regulated within one quarter of one percent. The test voltage shall be as specified on the detail specification.

4.6.1.1 Initial candlepower. Lamps shall be photometered and the spherical candlepower recorded. The lamps shall be sufficiently seasoned (burned) to have reached stable values of current and candlepower before photometering. The test shall be conducted at the test voltage specified in the detail specification. Any individual lamp having an initial candlepower more or less than that specified in the applicable detail specification by more than the allowable tolerance shall be considered defective.

See detail specification for applicable acceptance criteria Code. For Code A if more than 20 percent of the test lamps are defective, the lamp type shall fail to qualify. For Code B if more than 10 percent of the test lamps asre defective, the lamp type shall fail to qualify.

4.6.1.2 Candles (spherical) per watt (efficacy). The efficacy test shall be conducted after lamp seasoning. Any lamp having an efficacy less than that specified in the detail specification shall be considered defective.

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See detail specification for applicable acceptance criteria Code. For Code A if more than 20 percent of the test lamps are defective, the lamp type shall fail to qualify. For Code B if more than 10 percent of the test lamps are defective, the lamp type shall fail to qualify.

4.6.1.3 Current test. The current drawn by each lamp shall be measured at the time of initial photometry.

See detail specification for applicable acceptance criteria Code. For Code A no more than 20 percent of the test lamps shall deviate from the specified current by more than the allowable tolerance or the lamp type shall fail to qualify. For Code B no more than 10 percent of the test lamps shall deviate from the specified current by more than the allowable tolerance or the lamp type shall fail to qualify.

4.6.1.4 Candlepower maintenance. The candlepower maintenance test shall be conducted at the point in the lamp's life specified in the detail specification. The test conditions and procedures shall be the same as for initial candlepower measurements. Where average candlepower maintenance is specified, the test sample shall have an average candlepower output at the maintenance point equal to or exceeding that specified in the detailed specification as modified by 4.6.1.4.1 or the lamp type shall fail to qualify. Where candlepower maintenance is by individual lamps, no more than 10 percent of the test lamps shall have a candlepower maintenance output of less than that specified in the detail specification or the lamp type shall fail to qualify.

4.6.1.4.1 Candlepower maintenance tolerance. Where average candlepower maintenance output is specified, the average maintenance specified in the detail specification shall be "modified" as follows:

Sample	Allowable variation in average candlepower output at maintenance in percent of initial candlepower
100 and above	2
40 - 99	4
20 - 39	6
10 - 19	8

4.6.2 Life test. Lamps shall be tested for compliance with paragraph 3.5.1. Lamps shall be operated on AC or DC voltage as specified in the detail specification sheet. For life testing, the voltage shall be maintained within ± 1/2 percent of the specified test voltage. The test voltage shall be as specified on the detail specification sheet.

4.6.2.1 Life test tolerance. Where average life is specified, the average life specified in the detail specification shall be "modified" as follows to take account of the statistical effect of sample size.

Sample size	Allowable percent variation from specified life
10 - 19	8
20 - 39	7
40 - 99	6
100 and above	5

4.6.3 Vibration. The lamp shall be installed in a fixture as required and then subjected to the vibration test of MIL-STD-810B, Procedure I, Curve Z, Parts 1, 2, 3; Curve B, Part 4.

4.6.4 Shock. The lamp shall be tested as specified in MIL-STD-810B, Procedure I, Half Sine Shock Pulse, Amplitude 15 gs, Duration 11 ms.

4.6.5 Humidity. The lamp shall be tested in accordance with the humidity test, Procedure I, of MIL-STD-810B. After this test, the lamp shall be visually inspected for bulb to base integrity and operation.

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4.6.6 Dimensional examination. Lamps shall be examined for the dimensions specified on the detail specification sheet. Lamps shall be sampled in accordance with MIL-STD-105D, inspection level S-3. The AQL for acceptance for dimensions having a tolerance other than axial alignment and light center length shall be 1.0 expressed as defects per hundred units. For axial alignment, the AQL shall be 10.0 expressed as percent defective. For light center length, the AQL shall be 4.0 expressed as percent defective. Where a maximum or minimum dimension is specified, no lamps will be accepted which have dimensions less than a specified minimum nor greater than a specified maximum.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Levels A and B. The lamps shall be packaged in a box conforming to PPP-B-566 or PPP-B-676 and mechanically protected using separators, trays, or cushioning. Not more than 25 lamps shall be packaged per unit package. The packaged lamps shall be intermediate packaged in a close-fitting box conforming to PPP-B-636, class domestic, style optional. The intermediate package shall contain no more than 500 lamps.

5.1.2 Level C. The lamps shall be packaged in accordance with the manufacturer's standard commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Lamps, packaged as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-591, class II; PPP-B-601, overseas type; PPP-B-621, class 2; PPP-B-636, grade V3c or V3s; PPP-B-640, class 2; or PPP-B-1364. The gross weight of wood and wood-cleated boxes shall not exceed 200 pounds. Sealed case liners conforming to MIL-L-10547; type I, II, or IV, grade B, class 2 shall be used. The box shall be closed, waterproof sealed, and strapped or reinforced in accordance with the appendix to the applicable box specification.

5.2.2 Level B. Lamps, packaged as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-636, class domestic, style optional. The box shall be closed in accordance with the appendix to PPP-B-636.

5.2.3 Level C. Lamps, packaged as specified in 5.1, shall be packed in containers that will assure carrier acceptance and safe delivery to destination in compliance with the Uniform Freight Classification rules or National Motor Freight Classification rules, as applicable.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2 as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to any special marking required by the contract or order, marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The lamps covered by this specification are intended for use in automotive, flashlight, aviation, indicator, and similar applications.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable detail specification.
(c) Responsibility for inspection if other than specified (see 4.1).
(d) Sampling if other than specified (see 4.5.1.1).
(e) Specified packaging level (see 5.1).
(f) Specified packing level (see 5.2).
(g) Marking required (see 5.3).

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User activity:

Navy - AS

Preparing activity:

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